

PE LNG: Foundation Of Solar Cell Jetty





1. Background

2. Objective

3. Scope & Design of change

3.1 Scope of work

3.2 Detail design

3.3 Document related to be updated / New document

4. Risk analysis

4.1 Risk Class Evaluation

4.2 Process Hazard Analysis (PHA)

4.3 ความจำเป็นในการ Update Pre incident Plan

4.4 กฎหมายที่เกี่ยวข้องกับ MOC

5. Budgetary & Schedule







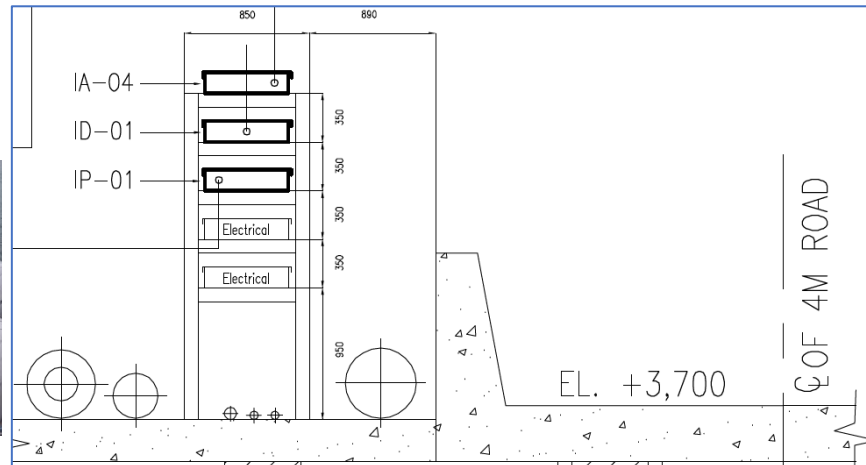
Solar PV is one of the key projects supporting the site's decarbonization plan and long-term Net Zero target.

Environmental Benefit of Solar PV : 1 kW of solar PV -----> reduce **901 kgCO₂/year** = **101 trees planted**

Jetty PV project: -----> **600 540.6 tCO₂/year** = **60,600 trees**

Process Area Limitation : Most available areas in the Process Area have already been allocated for future plant facilities.

Jetty Opportunity : The Jetty has a long continuous structure toward the sea with existing cable tray routes provide a clear linear reference for PV layout planning.



Objective



FROM	Connection Type	Target	Distance (m)
ISB Building	(JB-O3) 0300-WR-4-016A-JB	EL3	1,130 m
EL3	(JB-O3) 0300-WR-4-016B-JB	EL5	550 m
EL5	(JB-O3) 0300-WR-4-016C-JB	EL9	1136 m

Cable Size

2 Line parallel:

LV1-0300-WR-4-016-JB-01 & 016-JB-02

3P-4W-400V, 4C-400+1C-240

ID=77.5 mm, OD=93 mm, 052309 5MVSP

Electrical System Capacity: AC Conceptual Calculation



Regulatory Standard: BS 5467, IEC/EN 60502-1, IEC/EN 60228 with 0.85 factor



Cable Specification: 2 x Parallel 400 mm² 5MVSP



Installation: Perforated Cable Tray without cover (Method E)



Voltage Rating: 0.6/1(1.2) kV Rated Low Voltage



LOW VOLTAGE CABLES

0.6/1(1.2) kV CV-SWA (CU/XLPE/PVC/SWA/PVC)
FOUR CORES WITH GROUND CROSS-LINKED POLYETHYLENE INSULATED
AND POLYVINYL CHLORIDE SHEATHED POWER CABLE WITH STEEL WIRE ARMOUR



Total Rated Current 1324 A
(Load used 63A)

Number x Size of Conductor	Nominal Cross -sectional Area		Conductor				Nominal Thickness of Insulation		Approx. Thickness of Inner Covering	Diameter over Inner Covering (Approx.)	Nominal Diameter of Steel wire armour	Nominal Thickness of Overheath	Overall Diameter (Approx.)	Cable Weight (Approx.)	Stand- off Packing
			Minimum Number of Wires		Diameter (Approx.)										
	Phase	Ground	Phase	Ground	Phase	Ground									
							mm ²	mm ²							
4x1.5/1.5	1.5	1.5	7	7	1.6	1.6	0.7	0.7	1.0	10.6	0.8	1.8	17	450	1,000
4x2.5/2.5	2.5	2.5	7	7	2.0	2.0	0.7	0.7	1.0	11.8	0.8	1.8	18	530	1,000
4x4/4	4	4	7	7	2.5	2.5	0.7	0.7	1.0	13.1	1.25	1.8	21	780	1,000
4x6/6	6	6	7	7	3.1	3.1	0.7	0.7	1.0	14.6	1.25	1.8	22	940	1,000
4x10/10	10	10	6	6	3.7	3.7	0.7	0.7	1.0	16.4	1.25	1.8	24	1,200	1,000
4x16/16	16	16	6	6	4.6	4.6	0.7	0.7	1.0	18.8	1.6	1.8	27	1,720	1,000
4x25/16	25	16	6	6	5.9	4.6	0.9	0.7	1.0	23.3	1.6	1.8	31	2,295	1,000
4x35/16	35	16	6	6	7.0	4.6	0.9	0.7	1.0	26.2	1.6	1.9	35	2,820	500
4x50/25	50	25	6	6	8.1	5.9	1.0	0.9	1.2	30.3	2.0	2.1	40	3,930	500
4x70/35	70	35	12	6	9.6	7.0	1.1	0.9	1.2	34.8	2.0	2.2	44	5,080	500
4x95/50	95	50	15	6	11.4	8.1	1.1	1.0	1.2	39.8	2.0	2.4	50	6,530	300
4x120/70	120	70	18	12	12.7	9.6	1.2	1.1	1.4	44.3	2.5	2.5	55	8,450	300
4x150/95	150	95	18	15	14.1	11.4	1.4	1.1	1.4	49.2	2.5	2.7	61	10,140	200
4x185/95	185	95	30	15	15.8	11.4	1.6	1.1	1.6	55.2	2.5	2.9	67	12,165	200
4x240/120	240	120	34	18	18.1	12.7	1.7	1.2	1.6	61.9	2.5	3.1	74	15,170	200
4x300/150	300	150	34	18	20.4	14.1	1.8	1.4	1.6	68.8	2.5	3.3	82	18,365	200
4x400/240	400	240	53	34	23.5	18.1	2.0	1.7	1.8	78.7	3.15	3.7	94	24,885	200

Total System power Capacity
733.72 kW ที่ pf 0.8
EIT 022001-64

Number x Size of Conductor	Maximum DC Resistance of Conductor at 20°C		Minimum Insulation Resistance at 20°C		AC Resistance at 90°C		Inductive Reactance		Current Carrying Capacity	
	Phase	Ground	Phase	Ground	Ω/km	Ω/km	Ω/km	Ω/km	On Cable Tray at 40°C	Direct Burial in Ground at 30°C
4x1.5/1.5	12.1	12.1	1,002	1,002	15.43	0.1105	23	27		
4x2.5/2.5	7.41	7.41	839	839	9.45	0.1040	30	36		
4x4/4	4.61	4.61	702	702	5.88	0.0986	40	47		
4x6/6	3.08	3.08	594	594	3.93	0.0944	51	57		
4x10/10	1.83	1.83	509	509	2.33	0.0866	71	76		
4x16/16	1.15	1.15	421	421	1.47	0.0831	90	98		
4x25/16	0.727	1.15	427	421	0.927	0.0833	119	125		
4x35/16	0.524	1.15	366	421	0.669	0.0809	147	150		
4x50/25	0.387	0.727	351	427	0.494	0.0803	179	176		
4x70/35	0.268	0.524	330	366	0.343	0.0795	228	217		
4x95/50	0.193	0.387	281	351	0.247	0.0776	277	256		
4x120/70	0.153	0.268	276	330	0.197	0.0774	321	290		
4x150/95	0.124	0.193	288	281	0.160	0.0779	369	327		
4x185/95	0.0991	0.193	294	281	0.129	0.0781	421	366		
4x240/120	0.0754	0.153	275	276	0.0991	0.0773	497	422		
4x300/150	0.0601	0.124	259	288	0.0803	0.0767	571	475		
4x400/240	0.0470	0.0754	251	275	0.0646	0.0764	662	575		

Electrical System Voltage Drop: AC Conceptual Calculation

Voltage Drop Analysis (EIT 022013-22) → Requirement: VD < 5% from Inverter to MDB(LOAD)

Cal use length = 1130m
(EL3 shortest distance)



VD Limitation: Calculation shows the cable reaches 5% VD limit at **316.36 A Total.**



Actual Performance: Effective Capacity for PV system (pf = 1.0 IEEE 1547, IEC 61727, PEA / MEA) is constrained to **219.18 kW.**

Electrical System Capacity: System Design Optimization

PHASE 1: IMMEDIATE EXECUTION

200kW

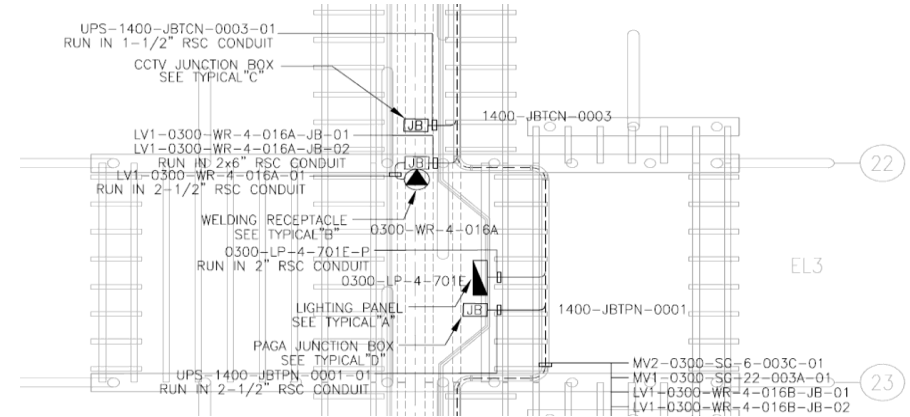
TOTAL AC CAPACITY

288.8A

INVERTER RATED
CURRENT OUTPUT

4.87%

CALCULATED AC
Voltage Drop



Strategic Placement: Inverters and SMDB installed at **EL3**.



Connection Point: Proximity to **JB-016A** significantly reduces AC path length.



VD Mitigation: Shorter cable runs ensure lower impedance and maximum power throughput.



Stability: Critical for maintaining performance at the load-side.

Alternative: EL5/EL9: Connecting to 016B-JB or 016C-JB results Higher VD leads to significant yield losses and potential system trips.



Electrical System Capacity: System Design Optimization

PHASE 1.5: FUTURE 100 KW EXPANSION

100kW

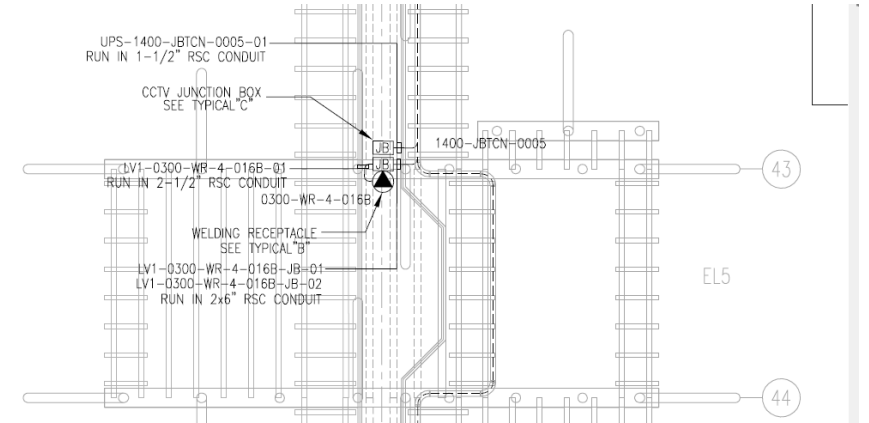
TOTAL AC CAPACITY

144.4A

Total INVERTER RATED
CURRENT OUTPUT

4.29%

Total CALCULATED
AC Voltage Drop



Strategic Placement: Inverters and SMDB installed at **EL5**.



Connection Point: Proximity to **JB-016B** significantly reduces AC path length.



New Parallel Line: requires **Dual Parallel Cables** (2x 4C-400 mm²) from **ISB** → **EL5**



Jumper Concept: Optimized single-run output for short-distance link to **JB-EL5**

Alternative: EL5/EL9: Connecting to 016B-JB or 016C-JB results Higher VD leads to significant yield losses and potential system trips.



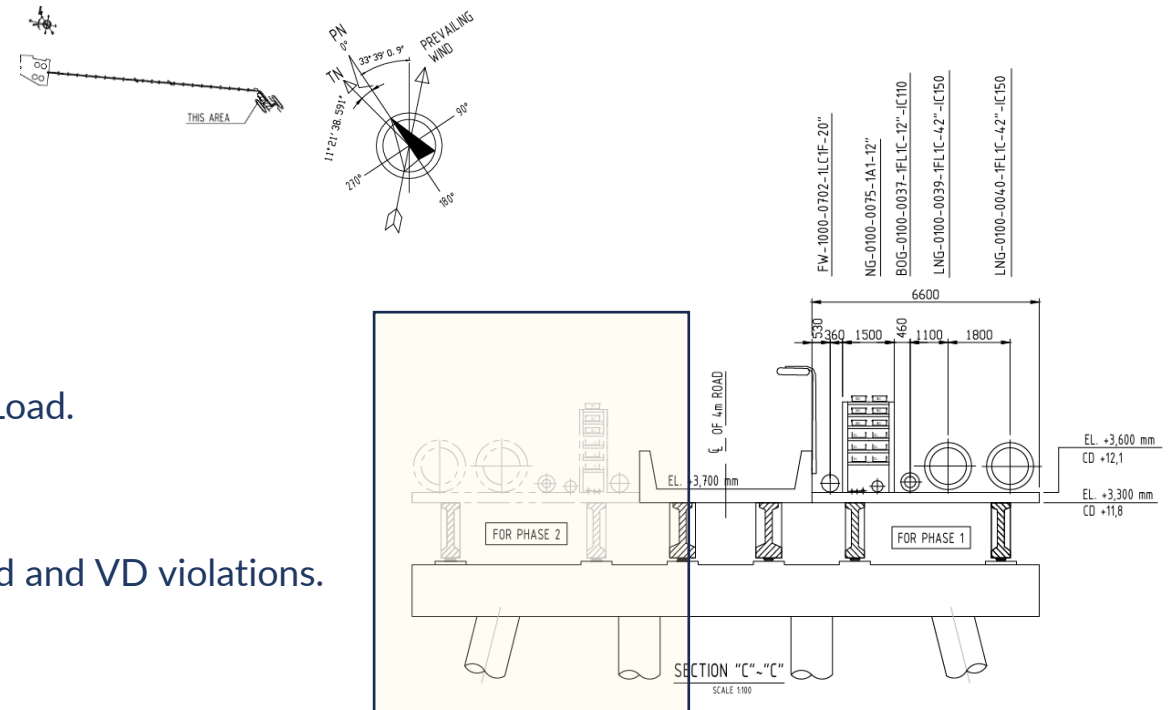


Detail design

Electrical System Capacity: System Design Optimization

PHASE 2: FUTURE 300KW EXPANSION

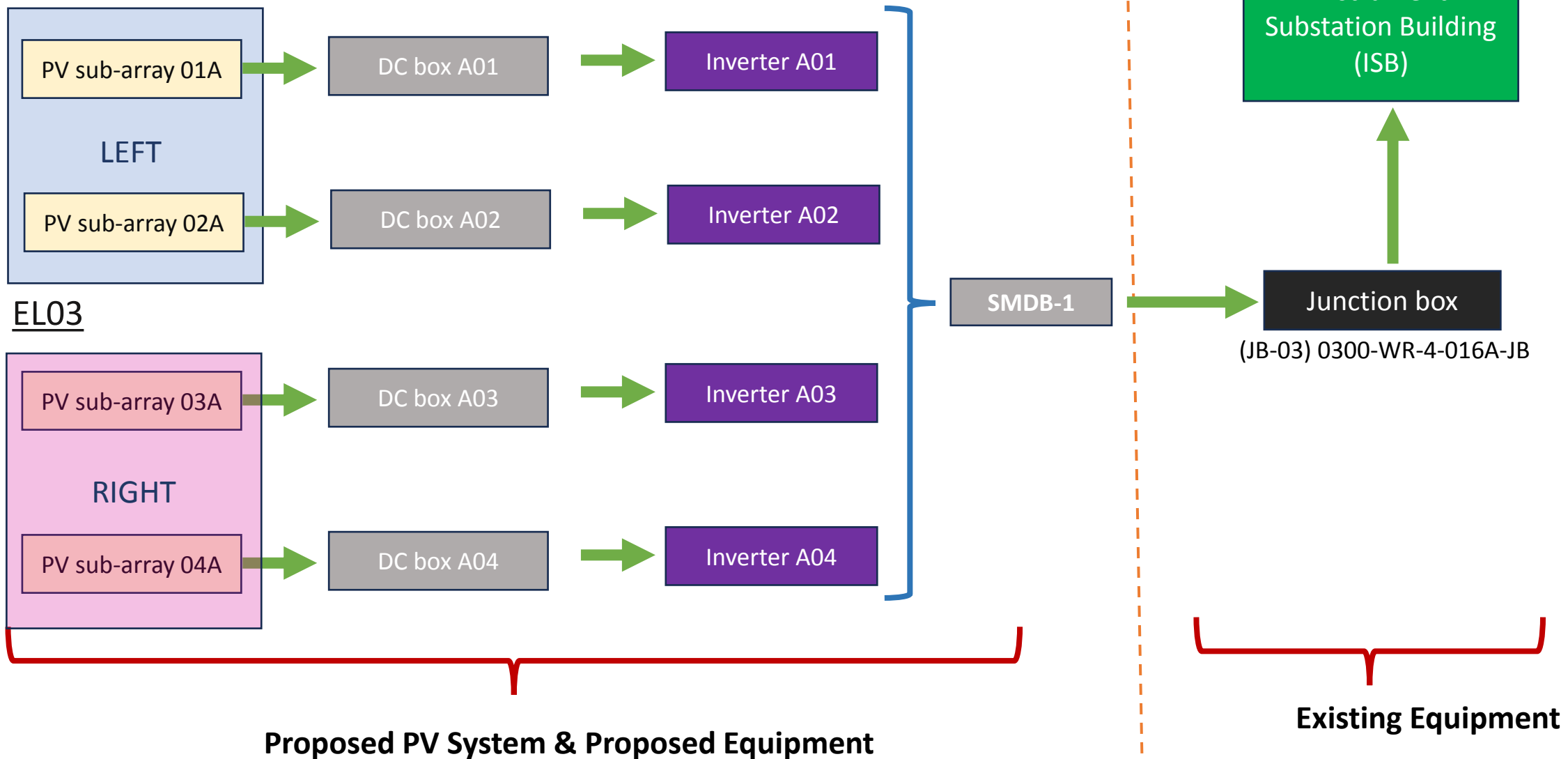
- + **Capacity Scaling:** Future expansion target of +300 kW.
- 🏗️ **Infrastructure:** Requires new 400mm² (or larger) AC circuits.
- 📦 **New Distribution:** Construction of a dedicated Junction Box near ISB Load.
- ⊘ **Constraint Note:** Existing JB's cannot be reused due to current overload and VD violations.





Detail design For Phase1

Electrical System Concept PV SYSTEM DIAGRAM





Detail design For Phase1

Preliminary PV System Specifications

PV module: Trina Solar Vertex N TSM-NEG21C.20 715W , total 320 modules length 2.384 m.

PV optimizer: HUAWEI MERC-1300W-P 2:1 , total 160 optimizers

For PV sub-array:

PV 01A.L – 4 strings arranged in sequential index order with 10, 10, 10, 10 optimizers per string.

PV 02A.L – 4 strings arranged in sequential index order with 10, 10, 10, 10 optimizers per string.

PV 03A.R – 4 strings arranged in sequential index order with 10, 10, 10, 10 optimizers per string.

PV 04A.R – 4 strings arranged in sequential index order with 10, 10, 10, 10 optimizers per string.

Inverter: HUAWEI sun2000-50kti-m3 , total 4 inverter for PV A

NOTES: PV system sizing, inverter selection, inverter location, and PV-side design are based on the following criteria:

DC Voltage Drop: Total DC voltage drop from PV modules to inverter shall be $< 3\%$ of V_m at STC.

AC Voltage Drop: Total AC voltage drop from the receiving point to the final load shall be $< 5\%$.

Optimizer String Length: Each string shall have more than 8 MERC-1300W optimizers.

DC/AC Ratio: The inverter DC/AC ratio shall be maintained at approximately 1.14–1.17.

String Power Balance: Input power difference between strings shall not exceed 2 kW.



Detail design For Phase1

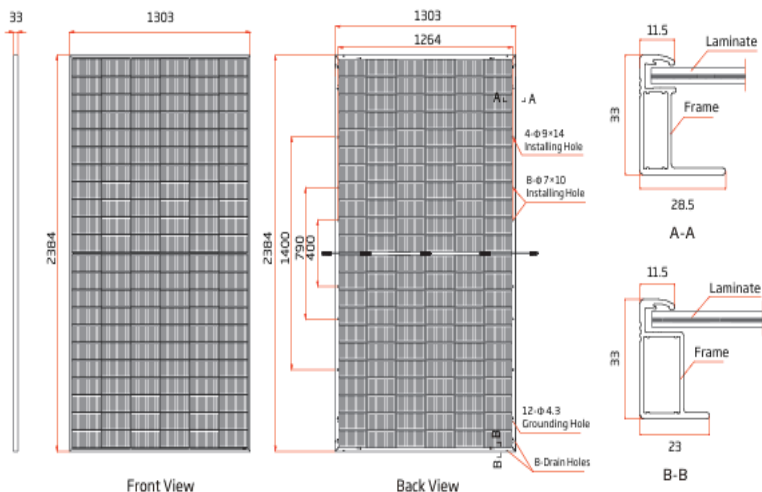
Preliminary PV System Specifications: Equipment Datasheets

Solar Cells	N-type i-TOPCon Monocrystalline
No. of cells	132 cells
Module Dimensions	2384×1303×33 mm (93.86×51.30×1.30 inches)
Weight	38.3 kg (84.4 lb)
Front Glass	2.0 mm (0.08 inches), AR Coating Heat Strengthened Glass
Back Glass	2.0 mm (0.08 inches), Heat Strengthened Glass (white Coating)
Frame	33mm (1.30 inches) Anodized Aluminium Alloy
J-Box	IP 68 rated
Cables	Photovoltaic Technology Cable 4.0mm ² (0.006 inches ²) Portrait: 350/280 mm (13.78/11.02 inches) Length can be customized
Connector	MC4 EV02 / TS4 Plus / TS4*
Packaging	Modules per box: 33 pieces Modules per 40' container: 594 pieces

*Please refer to regional datasheet for specified connector.

Testing Condition	STC	NOCT	BNPI	STC	NOCT	BNPI
Peak Power Watts- $P_{MAX}(W_p)^*$	710	543	787	715	547	792
Power Selection (W)**	0 ~ +5					
Maximum Power Voltage- V_{MPP} (V)	40.9	38.5	40.9	41.1	38.7	41.1
Maximum Power Current- I_{MPP} (A)	17.36	14.12	19.23	17.40	14.14	19.28
Open Circuit Voltage- V_{oc} (V)	49.0	46.5	49.0	49.2	46.7	49.2
Short Circuit Current- I_{sc} (A)	18.40	14.83	20.39	18.44	14.86	20.43
Module Efficiency η_m (%)	22.9			23.0		

Backside Power Gain	5%	10%
Peak Power Watts- $P_{MAX}(W_p)$	751	787
Maximum Power Voltage- V_{MPP} (V)	41.1	41.1
Maximum Power Current- I_{MPP} (A)	18.27	19.14
Open Circuit Voltage- V_{oc} (V)	49.2	49.2
Short Circuit Current- I_{sc} (A)	19.36	20.28



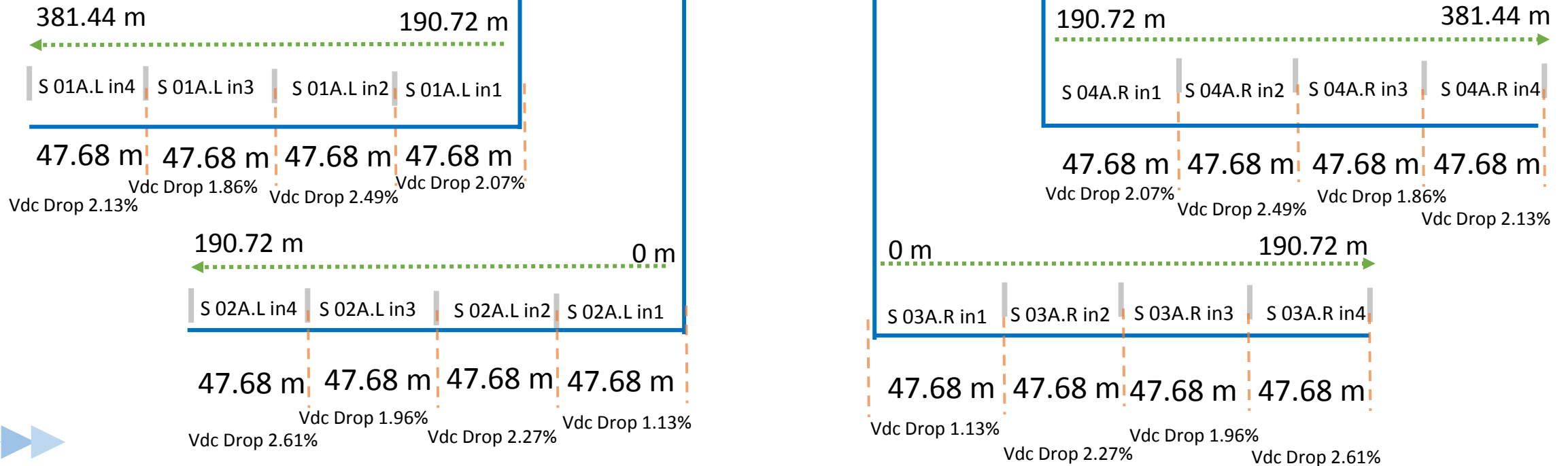
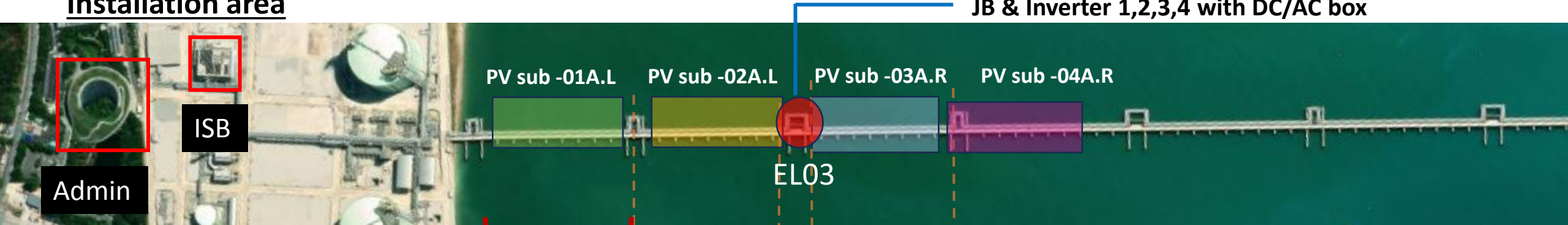
Technical Specifications	SUN2000-50KTL
Max. Efficiency	99%
European Efficiency	98.6%
Max. DC Usable Power	51,500 W
Max. Input Voltage	1,100 V
Max. Current per MPPT	22 A
Max. Short Circuit Current per MPPT	30 A
Min. Operating Voltage / Start Input Voltage	200 V / 250 V
Full Power MPPT Voltage Range	580 V ~ 850 V
MPPT Operating Voltage Range	200 V ~ 1000 V
Rated Input Voltage	720 V
Max. Number of Inputs	8
Number of MPP Trackers	4
Rated AC Active Power	46,000 W
Max. AC Apparent Power	50,500 VA
Max. AC Active Power ($\cos\phi=1$)	50,500 W
Rated Output Voltage	277V / 480V, 3W+PE
Rated AC Grid Frequency	50 Hz / 60 Hz
Max. Output Current	60.8 A
Adjustable Power Factor	0.8 LG ... 0.8 LD
Max. Total Harmonic Distortion	< 3%
Input-side Disconnection Device	Yes
Anti-Islanding Protection	Yes
DC Reverse-Polarity Protection	Yes
PV-array String Fault Monitoring	Yes
DC Surge Arrester	Type II
AC Surge Arrester	Type II
Insulation Detection	Yes
Residual Current Monitoring Unit	Yes

Technical Specification	MERC-1100W-P	MERC-1300W-P		
Input				
Rated Input DC Power	1100 W	1300 W		
Max. Input DC Power	1155W ¹	1430W ¹		
Max. input voltage	125 V			
MPPT operating voltage range	12.5 ~ 105 V			
Max. short-circuit current (Isc)	20 A			
Max. efficiency	99.5 %			
Weighted efficiency	99.0 %			
Overvoltage category	II			
Output				
Rated Output DC Power	1100W	1300W		
Max. Output DC Power	1155W	1430W ²		
Max. output voltage	80 V			
Max. output current	22 A			
Output bypass ³	Yes			
Shutdown output voltage per optimizer ⁴	1 V			
Standards Compliance				
Safety	IEC62109-1 (class II safety)			
RoHS	Yes			
General Data				
Dimension (W x H x D)	149 mm x 104 mm x 49 mm (5.9 in. x 4.1 in. x 1.9 in.)			
Weight (including cables)	1.0 kg (2.2 lb.)			
Installation part (optional)	PV Module Frame Plate/T-shaped Bolt ⁵			
Input connector	Staubli MC4			
Input wire length	0.1 m (short input cable version) ⁶			
Output connector	Staubli MC4			
Output wire length	0.1 m (+), 5.1 m (-) (short input cable version) ⁶			
Operating temperature/humidity range	-40°C to +85°C / 0%-100% RH			
Degree of protection	IP68			
Compatible Inverter	SUN2000-8/10/12/15/17/20KTL-M2 SUN2000-30/36/40KTL-M3 SUN2000-12/15/17/20/23/25KTL-M5 SUN2000-50KTL-M3			
String Configuration (Full Optimizer Configuration) ^{5/9/10} ¹ MERC-1100/1300W-P support full optimizer configuration only				
	SUN2000-8-20KTL-M2	SUN2000-12-25KTL-M5	SUN2000-30-40KTL-M3	SUN2000-50KTL-M3
Minimum optimizers per string	8	8	8	8
Maximum optimizers per string	25	25	25	20
Maximum DC power per string	20,000 W	20,000 W	20,000 W	20,000 W



Detail design For Phase1

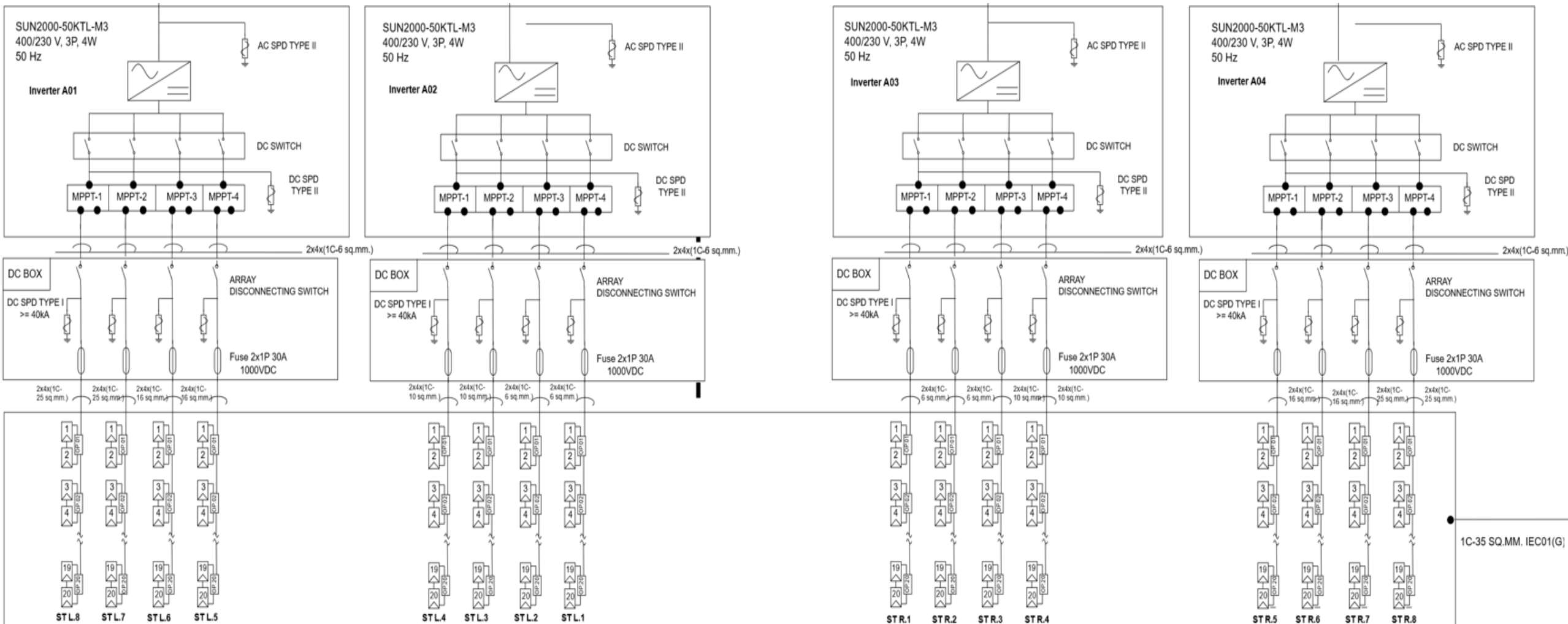
Installation area





Detail design For Phase1

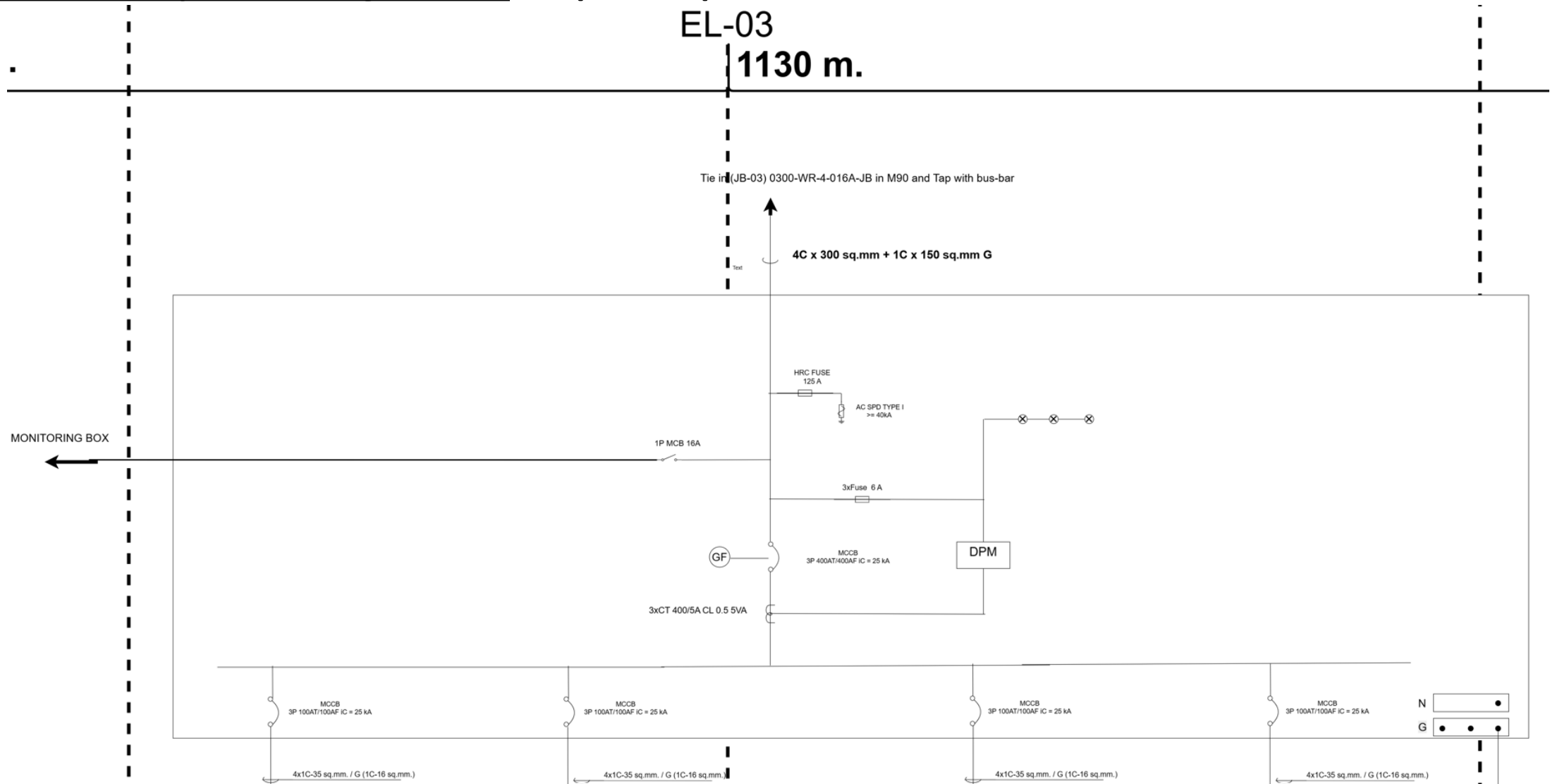
PV Electrical System Configuration: SLD part DC phase1





Detail design For Phase1

PV Electrical System Configuration: SLD part AC phase 1

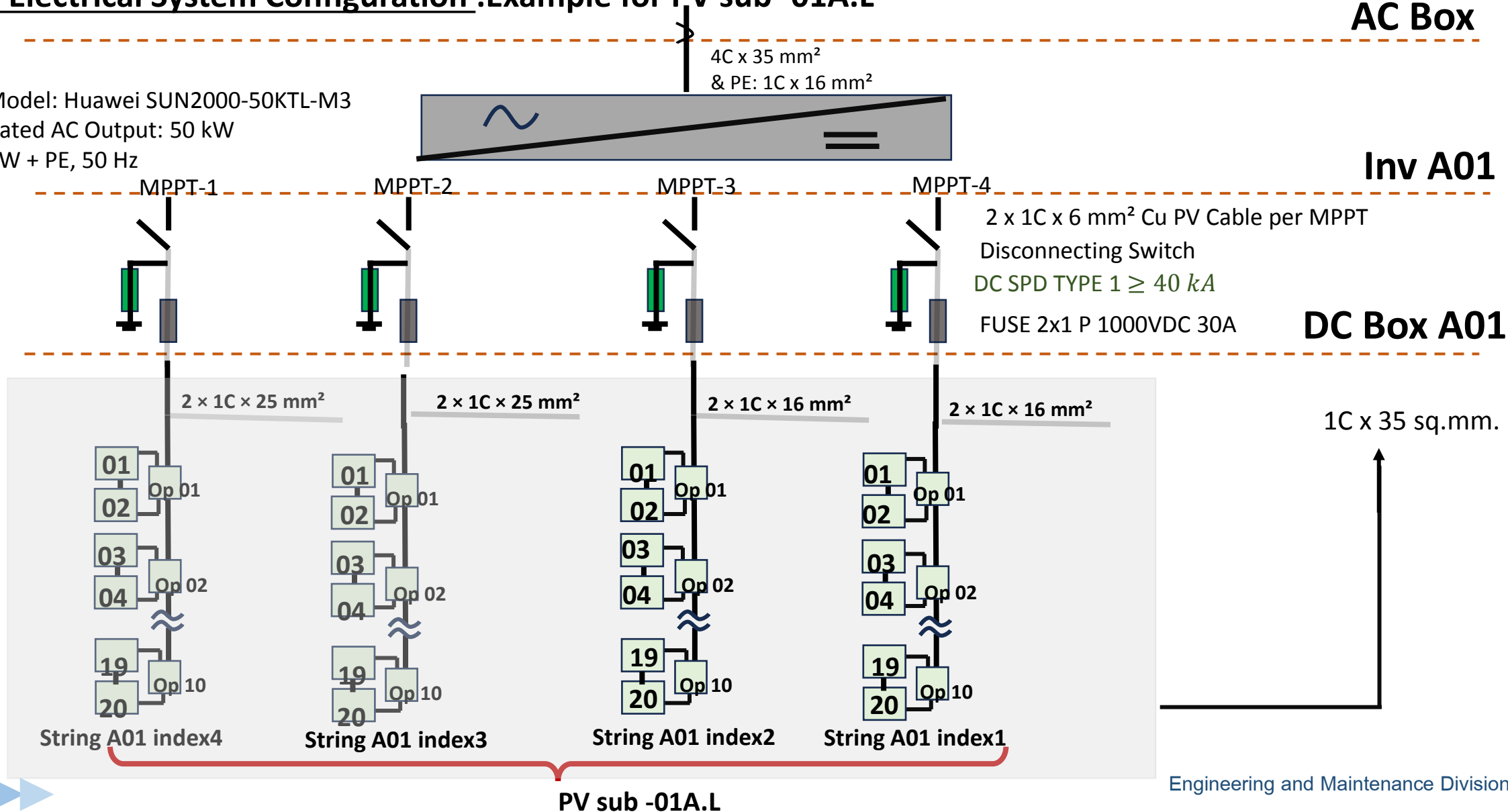




Detail design For Phase1

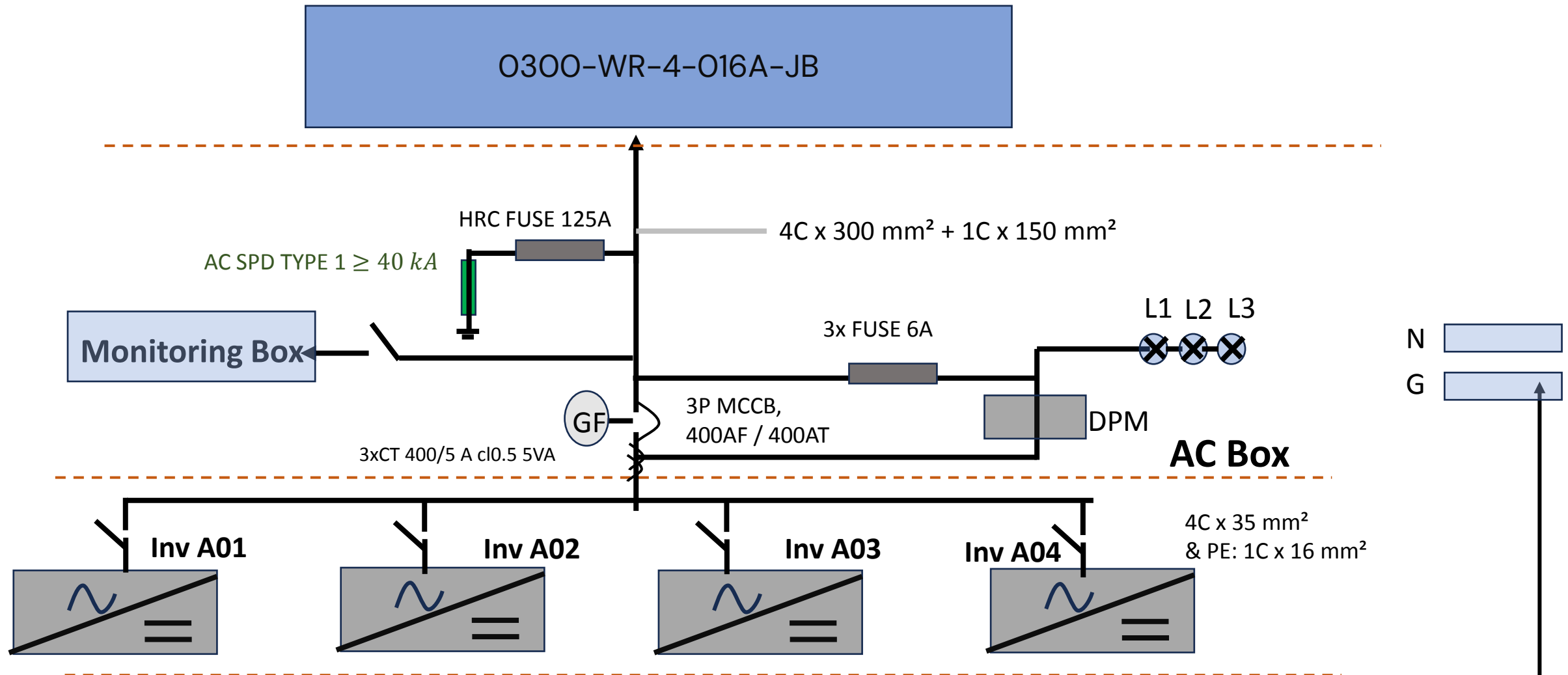
PV Electrical System Configuration :Example for PV sub -01A.L

Model: Huawei SUN2000-50KTL-M3
Rated AC Output: 50 kW
3W + PE, 50 Hz





PV Electrical System Configuration :JB / AC part



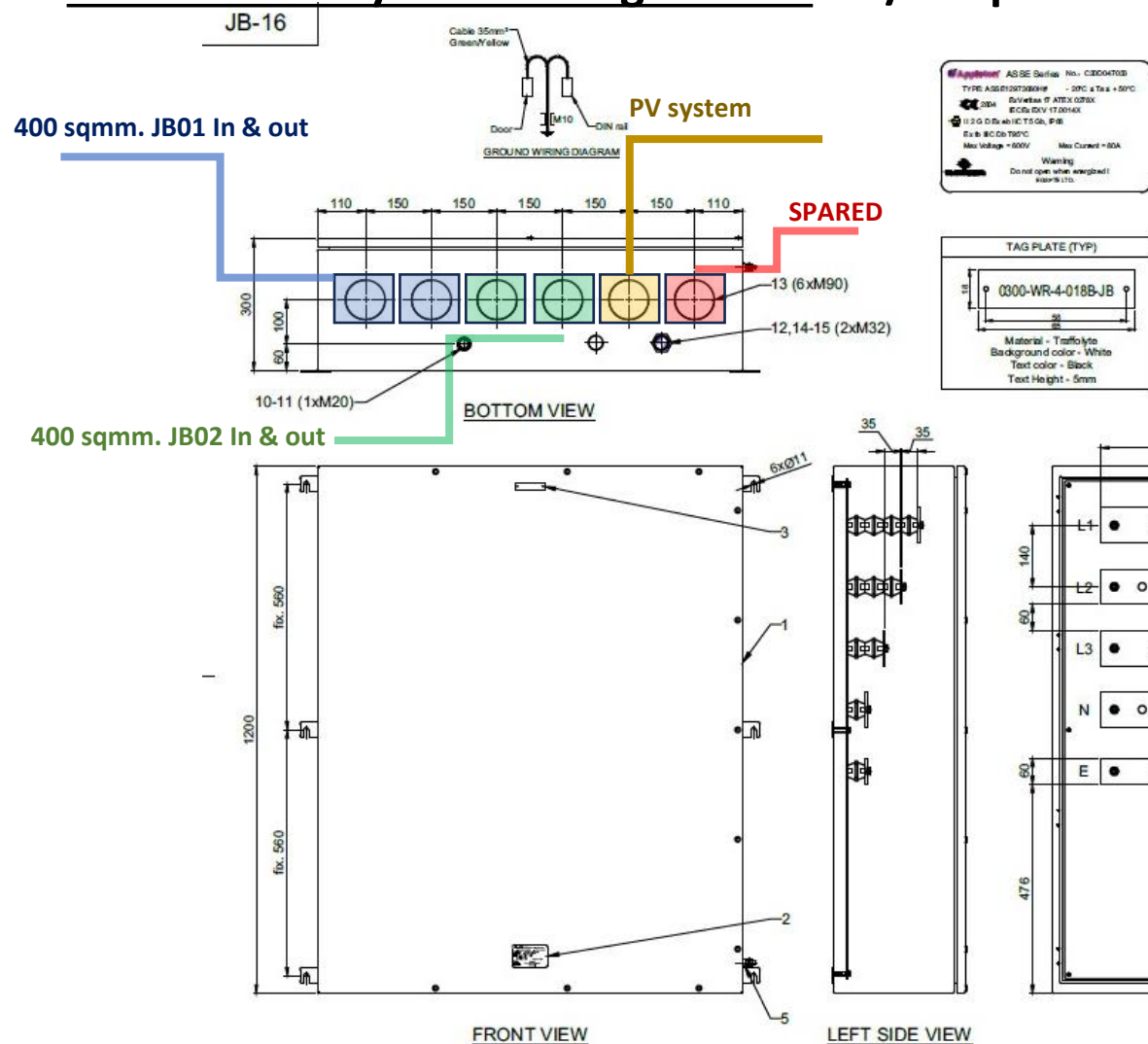


Detail design For Phase1

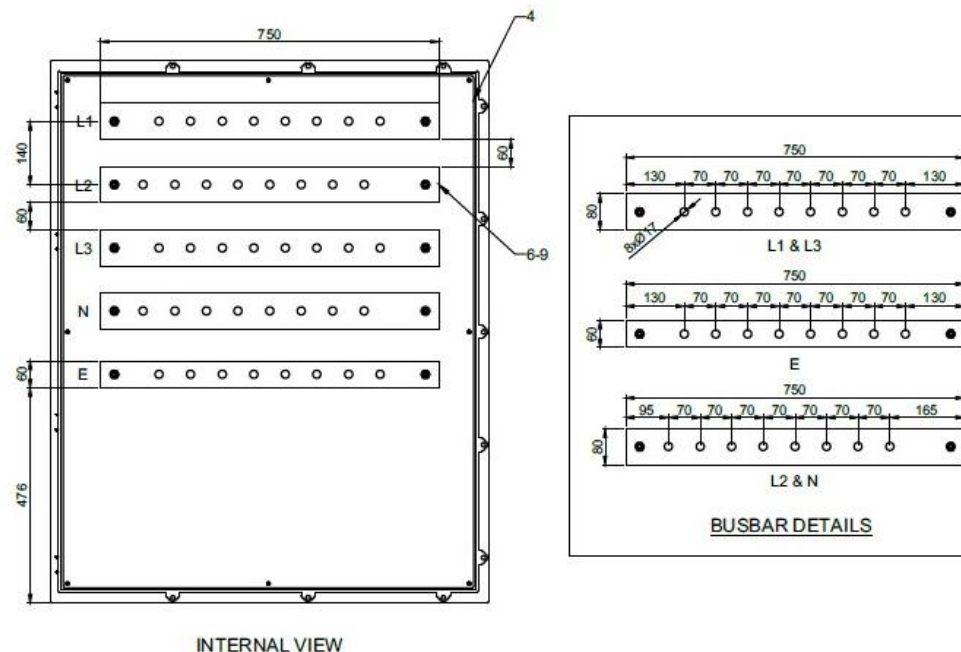
PV Electrical System Configuration: SLD



PV Electrical System Configuration :JB / AC part



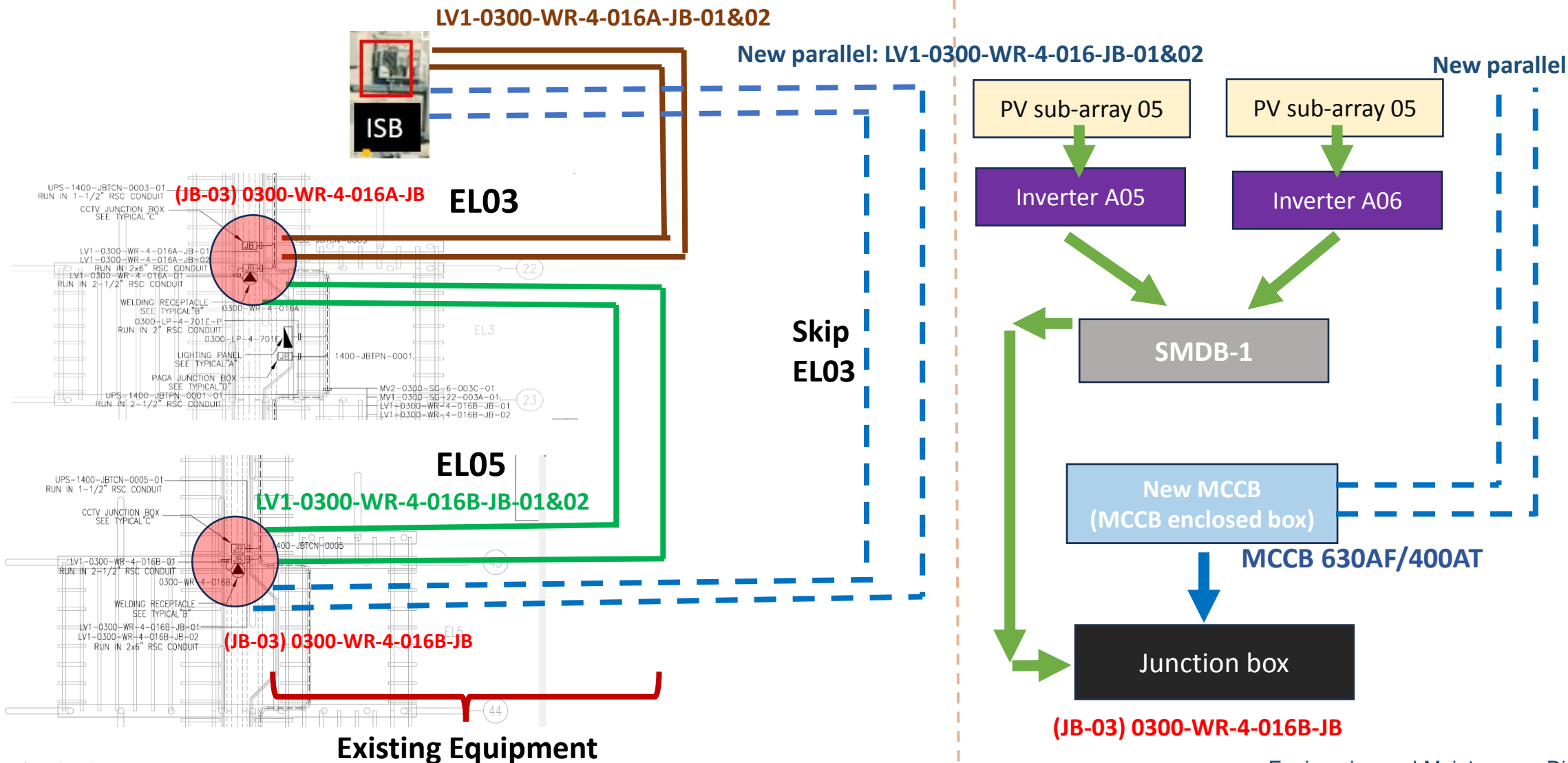
SL. NO.	QTY	MATERIAL	CATALOG REF	MAKE
1	1	Enclosure in SS316L with 1.5mm thick Dimensions : 1200(L) x 970(W) x 300(D)mm	ASSE12973060H	Emerson
2	1	Ex. Label in adhesive type polyester	-	-
3	1	Tag plate in Traffolyte (65x18x1.5mm)	-	-
4	1	Internal mounting plate Zinc plated steel	-	-
5	1	M10 Earth stud (earth crossing)	-	-
6	4	Copper Busbar [80(W)x5(T)x750(L)mm]	-	-
7	1	Earth bar size [60(W)x5(T)x750(L)mm]	-	-
8	22	Insulators	-	-
9	2	DiN rail for cable support	-	-
10	1	M20 Breathe and drain in SS316	DBE20S	Emerson
11	1	M20 Clearance entry (Ø20.5)	-	-
12	2	M32 Clearance entry (Ø32.5)	-	-
13	6	M90 Clearance entry (Ø90.5)	-	-
14	1	M32 Blanking plug in Nickel plated brass	757DM45	A.T.X
15	1	Sealing washer and locknut	32ETS2+095703	A.T.X





Detail design For Phase1.5

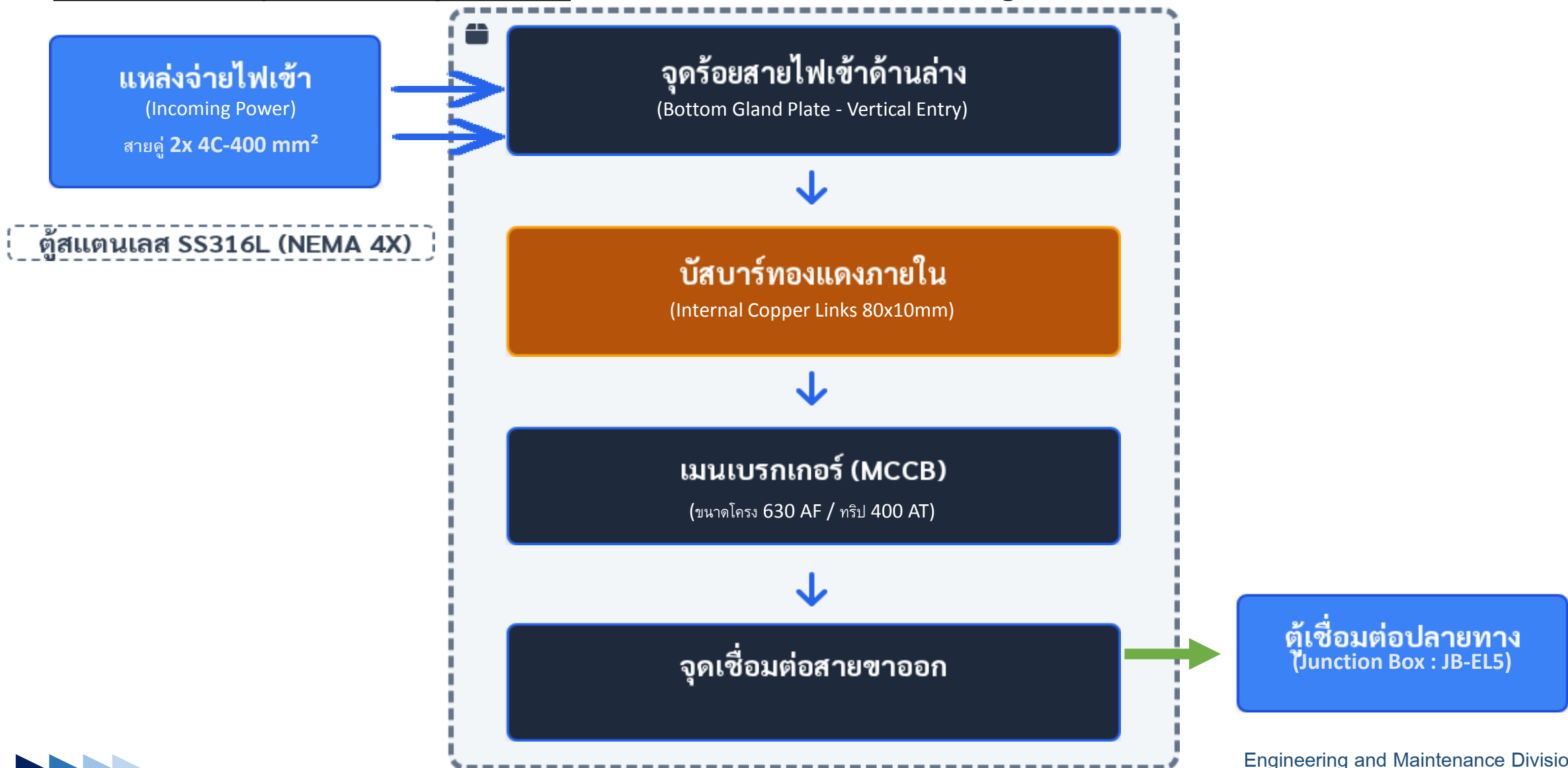
PV Electrical System Configuration :Conceptual in the near future





Detail design For Phase1.5

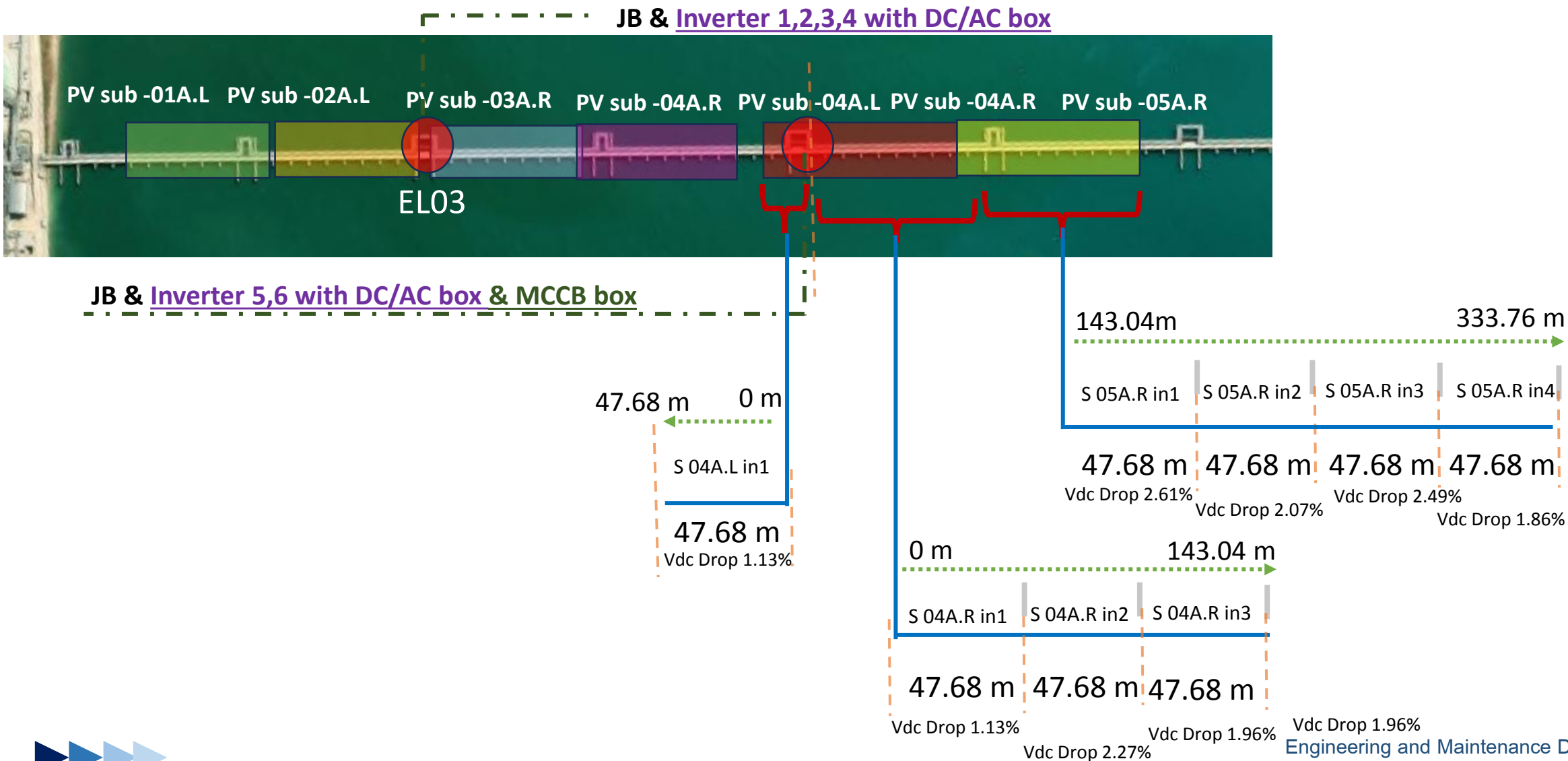
PV Electrical System Configuration :New MCCB Enclosed box design





Detail design For Phase1.5

PV Electrical System Configuration :Installation area1&1.5

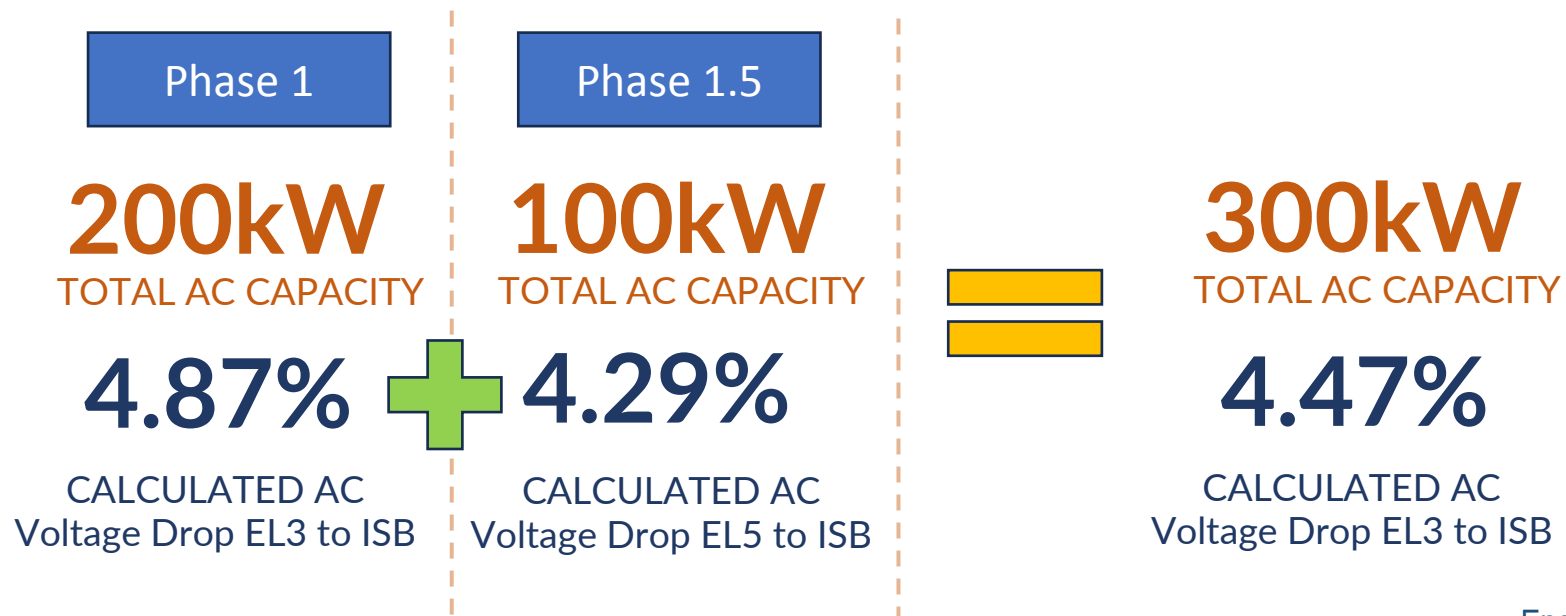
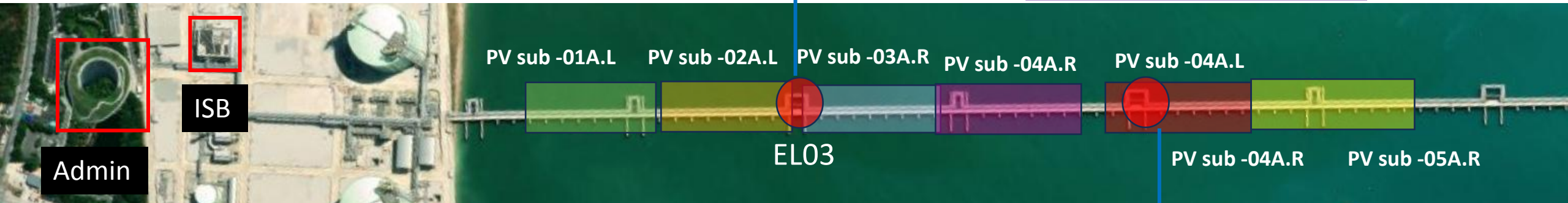




Detail design For Phase1.5

PV Electrical System Configuration :Summary of Phase 1&1.5

Installation area





Main scope

1. Solar PV System

- a. Installed capacity: 228.8kW DC / 200 kW AC with PV Module: Trina 715 W × 320 modules
- b. Optimizer: Huawei MERC-1300W-P × 160 units

2. PV String Configuration

- a. PV 01A: 4 strings & PV 02A: 4 strings, PV 03A: 4 strings & PV 04A: 4 strings
- b. String length: 10 Optimizers/String & Module/String: 20 Modules/String

3. PV Mounting Structure

- a. Structure suitable for Jetty / marine environment
- b. Corrosion-resistant material and fasteners
- c. PV frame bonding and earthing connection

4. Inverter Installation

- a. Huawei SUN2000-50KTL-M3: 4 set
- b. Total inverter capacity: 200 kW AC , DC/AC Ratio: approximately 1.144

5. DC System

- a. DC cable: 1C × 6, 10, 16,25 mm²
- b. DC isolator, gPV Fuse, PV DC SPD
- c. DC Voltage Drop: < 3% of Module Vmp with Maximum DC voltage: < 1,100 Vdc





Main scope

6. AC Box A — 50 kW Inverter

- a. MCCB: 3P, 400AF / 400AT & AC SPD Type 1+2 or Type 2
- b. CT, Digital Power Meter, Neutral Bar, Earth Bar & AC cable+ PE

7. AC Connection to MDB / PCC

- a. Connect AC Box A and B to existing MDB / PCC
- b. Verify Voltage Rise / Voltage Drop, Target AC Voltage Rise: $\leq 5\%$
- c. Confirm Grid Protection Setting before commissioning

8. Grounding and Lightning Protection

- a. Ground PV frame, structure, inverter, DC Box, AC Box, cable tray
- b. Install PE cable, earth bar, bonding jumper, test point, DC SPD and AC SPD
- c. Earth continuity and earth resistance test

9. Cable Works

- a. Cable pulling, gland, lug, termination, labeling, Cable tray extension and support modification
- b. Separate DC and AC cable routes with Outdoor / marine-grade cable accessories





Main scope

10. Monitoring and Control

- a. Huawei SmartLogger / Monitoring System
- b. RS485 / Ethernet communication
- c. PCC Meter and CT for export monitoring
- d. Inverter Alarm, Trip, SPD Fault, Communication Fault

11. Protection and Grid Complianc

- a. Anti-islanding protection
- b. Over/Under Voltage Protection & Over/Under Frequency Protection
- c. Harmonic measurement at PCC ,THDi / THDv and individual harmonic verification

12. Testing and Commissioning

- a. FAT / SAT
- b. DC polarity, Voc, insulation resistance, continuity test
- c. Grounding and earth resistance test ,Inverter startup and full-load test, Anti-islanding and protection trip test ,Voltage Rise / Drop and Power Quality Test

13. Training and Documentation

- a. Operation and Maintenance Training
- b. AS-Built Drawing & SLD, Layout, Cable Schedule, String Schedule
- d. Test Report, Commissioning Report, Datasheet, Warranty Documents



Cable schedule

No,	Cable TAG	Cable Status	Type	Voltage/Signal	FROM (Existing)	From (New/Add)	TO (Existing)	To (New/Add)	Length	EQUIP.	UNIT	SIZE	TYPE
1	C-0100-PM-001C-VSD-01	New	Control	DI/DO		VSD LP1C		LB04	TBC			10C-1.5	5MVNC
2	C-0100-PM-001C-VSD-02	New	Control	DI/DO		VSD LP1C		LB04	TBC			10C-1.5	5MVNC
3	C-0100-PM-002C-VSD-01	New	Control	DI/DO		VSD LP2C		LB12	TBC			10C-1.5	5MVNC
4	C-0100-PM-002C-VSD-02	New	Control	DI/DO		VSD LP2C		LB12	TBC			10C-1.5	5MVNC
5	LV1-0100-PM-001C-VSD-01	New	LV Power	400V		LV Supply		VSD LP1C	TBC			3C-10+1C-10	5MVSP
6	LV1-0100-PM-002C-VSD-01	New	LV Power	400V		LV Supply		VSD LP2C	TBC			3C-10+1C-10	5MVSP
7	LV2-0100-PM-001C-VSD-01	New	LV Power	230V		UPS Supply		VSD LP1C	TBC			2C-4+1C-4	5MVSP
8	LV2-0100-PM-002C-VSD-01	New	LV Power	230V		UPS Supply		VSD LP2C	TBC			2C-4+1C-4	5MVSP
9	MV3-0100-PM-001C-01	Existing	MV Power	6.6kV	MSS LP 1C	VSD LP1C	Motor LP 1C	Motor LP 1C	TBC		kW	3C-35	4MESP
10	MV3-0100-PM-002C-01	Existing	MV Power	6.6kV	MSS LP 2C	VSD LP2C	Motor LP 2C	Motor LP 2C	TBC		kW	3C-35	4MESP
11	MV3-0100-PM-001C-VSD-01	NEW	MV Power	6.6kV	MSS LP 1C	VSD LP2C	LB04	LB04	TBC		kW	3C-35	4MESP
12	MV3-0100-PM-002C-VSD-01	NEW	MV Power	6.6kV	MSS LP 2C	VSD LP2C	LB12	LB12	TBC		kW	3C-35	4MESP
13	S-0100-PM-001C-VSD-01	New	Control	AI/AO		VSD LP1C		DCS marshelling	TBC			4P-1.5	AN4
14	S-0100-PM-001C-VSD-02	New	Control	AI/AO		VSD LP1C		DCS marshelling	TBC			4P-1.5	AN4
15	S-0100-PM-002C-VSD-01	New	Control	AI/AO		VSD LP2C		DCS marshelling	TBC			4P-1.5	AN4
16	S-0100-PM-002C-VSD-02	New	Control	AI/AO		VSD LP2C		DCS marshelling	TBC			4P-1.5	AN4
17	UPS-0100-PM-001C-VSD-01	New	UPS	UPS		0300-U-002		VSD LP1C	TBC			2C-4+1C-4	5MVSP
18	UPS-0100-PM-002C-VSD-01	New	UPS	UPS		0300-U-002		VSD LP2C	TBC			2C-4+1C-4	5MVSP
19	UTP-VSD A / Fiber	New	-	Communication		VSD LP1C		CSCS Control cabinet	TBC			-	CAT.6A
20	UTP-VSD B / Fiber	New	-	Communication		VSD LP2C		CSCS Control cabinet	TBC			-	CAT.6A
21	Grounding for VSD LP 1A	New	Grounding	Grounding	Main ground loop	Main ground loop		VSD LP1C	TBC			>35sq.mm.	5GI
22	Grounding for VSD LP 2A	New	Grounding	Grounding	Main ground loop	Main ground loop		VSD LP2C	TBC			>35sq.mm.	5GI

Cable schedule

REV NO	SERIAL NUMBER	CABLE/CONDUIT NUMBER	FEEDER DESCRIPTION	FROM			TO			FOR	LENGTH (M)	EQUIP.	UNIT	VOLT	SIZE	TYPE	SIZE	GLAND SIZE	EQUIPMENT LOCATION DRAWING NO.	DRUM NO
		9									5656	RATING		PH-W-V	(IN.)		(SQ. MM)	SOURCE		
F1	1663	LV1-0300-WR-4-016A-JB-01	Jetty Trestle Welding Receptacle @ EL3	ISB	0300-PP-4-002WD-2		EL03	(JB-03) 0300-WR-4-016A-JB		Power	1130	43.5	KVA	3-4-400	6	5MVSP	4C-400+1C-240	ID=77.5 mm, OD=93 mm, 052309 / M90	1411	5MVSP-04-400-E-9999-013~015
F1	1664	LV1-0300-WR-4-016A-JB-02	Jetty Trestle Welding Receptacle @ EL3	ISB	0300-PP-4-002WD-2		EL03	(JB-03) 0300-WR-4-016A-JB		Power	1130	43.5	KVA	3-4-400	6	5MVSP	4C-400+1C-240	ID=77.5 mm, OD=93 mm, 052309 / M90	1411	5MVSP-04-400-E-9999-016~018
F1	1665	LV1-0300-WR-4-016A-01	Jetty Trestle Welding Receptacle @ EL3	EL03	0300-WR-4-016A-JB		EL03	0300-WR-4-016A		Power	8	43.5	KVA	3-4-400	2-1/2	5MVSP	4C-25+1C-16	ID=23.5 mm, OD=31.5 mm, 052303 / M32	1411	5MVSP-04-025-E-1311-008
F1	1666	LV1-0300-WR-4-016B-JB-01	Jetty Trestle Welding Receptacle @ EL5	EL03	0300-WR-4-016A-JB		EL05	(JB-03) 0300-WR-4-016B-JB		Power	550	43.5	KVA	3-4-400	6	5MVSP	4C-400+1C-240	ID=77.5 mm, OD=93 mm, 052309 / M90	1411	5MVSP-04-400-E-9999-033~034
F1	1667	LV1-0300-WR-4-016B-JB-02	Jetty Trestle Welding Receptacle @ EL5	EL03	0300-WR-4-016A-JB		EL05	(JB-03) 0300-WR-4-016B-JB		Power	550	43.5	KVA	3-4-400	6	5MVSP	4C-400+1C-240	ID=77.5 mm, OD=93 mm, 052309 / M90	1411	5MVSP-04-400-E-9999-035~036
F1	1668	LV1-0300-WR-4-016B-01	Jetty Trestle Welding Receptacle @ EL5	EL05	0300-WR-4-016B-JB		EL05	0300-WR-4-016B		Power	8	43.5	KVA	3-4-400	2-1/2	5MVSP	4C-25+1C-16	ID=23.5 mm, OD=31.5 mm, 052303 / M32	1411	5MVSP-04-025-E-1311-008
F1	1669	LV1-0300-WR-4-016C-JB-01	Jetty Trestle Welding Receptacle @ EL9	EL05	0300-WR-4-016B-JB		EL09	(JB-04) 0300-WR-4-016C-JB		Power	1136	43.5	KVA	3-4-400	6	5MVSP	4C-400+1C-240	ID=77.5 mm, OD=93 mm, 052309 / M90	1411	5MVSP-04-400-E-9999-007~009
F1	1670	LV1-0300-WR-4-016C-JB-02	Jetty Trestle Welding Receptacle @ EL9	EL09	0300-WR-4-016B-JB		EL09	(JB-04) 0300-WR-4-016C-JB		Power	1136	43.5	KVA	3-4-400	6	5MVSP	4C-400+1C-240	ID=77.5 mm, OD=93 mm, 052309 / M90	1411	5MVSP-04-400-E-9999-010~012
F1	1671	LV1-0300-WR-4-016C-01	Jetty Trestle Welding Receptacle @ EL9	EL09	0300-WR-4-016C-JB		EL09	0300-WR-4-016C		Power	8	43.5	KVA	3-4-400	2-1/2	5MVSP	4C-25+1C-16	ID=23.5 mm, OD=31.5 mm, 052303 / M32	1411	5MVSP-04-025-E-1311-008

Cable specification

5MVSC	XLPE INSULATED STEEL WIRE ARMORED PVC SHEATHED CABLE - MULTI CORE, CU/XLPE/PVC/SWA/PVC CABLE, FLAME RETARDANT(LSLH), VERMIN& RODENT PROOF,UV RESISTANT OVERSHEATH, 0.6/1(1.2) kV, CONTROL CABLE, IEC 60502
5MVSP	XLPE INSULATED STEEL WIRE ARMORED PVC SHEATHED CABLE - MULTI CORE, CU/XLPE/PVC/SWA/PVC+(PE) CABLE WITH PROTECTIVE EARTH CONDUCTOR, FLAME RETARDANT(LSLH), VERMIN& RODENT PROOF, UV RESISTANT OVERSHEATH, 0.6/1(1.2) kV, POWER CABLE, IEC 60502
4MESP	TR-XLPE INSULATED STEEL WIRE ARMORED PE SHEATHED CABLE - MULTI CORE, CU/TR-XLPE/SCT/PE/SWA/PE CABLE (WITH COPPER TAPE SCREENED, SEPARATOR TAPE SHALL BE SPUN-BOND AND OPTION SHALL BE WATER BLOCKING TAPE), FLAME RETARDANT(LSLH), VERMIN& RODENT PROOF,UV RESISTANT OVERSHEATH, 3.8/6.6(7.2) kV, POWER CABLE, IEC-60228
ANS	INSTRUMENT ELECTRONIC CABLE, CU/XLPE/PVC/SWA/PVC , STRANDED COPPER, XLPE INSULATION, 600V, TWISTED PAIR, ALUMINUM/SYNTHETIS POLYMER TAPE OVERALL & INDIVIDUAL SHIELD WITH DRAIN WIRE, PVC JACKET, ANTI-TERMITE AND RODENT, FLAME RETARDANT, UV RESISTANT
5GI	GROUNDING WIRE, CU/PVC , STRANDED COPPER,PVC INSULATION,600V,GREEN



Update document

Item	Document Name	Doc. Number	Revision
1	One line diagram 6.6 kV,0.4kV	TBR	
2	ISB Equipment plot plan	TBR	
3	ISB cable tray layout and cable route	TBR	
4	ICSS connection diagram	TBR	
5	SWGR logic diagram	TBR	
6	CSCS configuration diagram	TBR	

Add document

Item	Document Name	Doc. Number	Revision
1	VSD chilled water pump package.		





Priority Classification

Attachment1		
	Consequences	Likelihood
ผลกระทบด้านบุคลากร (Safety)	<u>Level 4</u> ทุพพลภาพหรือเสียชีวิตจากการปฏิบัติงาน	<u>Level 4</u> >1 ครั้งต่อปี
	<u>Level 3</u> เจ็บป่วยรุนแรงถึงขั้นหยุดงานตั้งแต่ 3 วันขึ้นไป	<u>Level 3</u> 1 ครั้งใน 3 ปี
	<u>Level 2</u> ต้องได้รับการรักษาทางการแพทย์ที่โรงพยาบาลถึงขั้นหยุดงานน้อยกว่า 3 วัน	<u>Level 2</u> 1 ครั้งใน 5 ปี
	<u>Level 1</u> ต้องได้รับการรักษาทางการแพทย์ที่โรงพยาบาลแต่ไม่หยุดงาน	<u>Level 1</u> 1 ครั้งใน 10 ปี
ผลกระทบด้านบุคลากร (Health)	<u>Level 4</u> ทุพพลภาพหรือเสียชีวิต	<u>Level 4</u> >1 ครั้งต่อปี
	<u>Level 3</u> ต้องได้รับการรักษาทางการแพทย์เพื่อฟื้นฟูถึงขั้นหยุดงาน	<u>Level 3</u> 1 ครั้งใน 3 ปี
	<u>Level 2</u> ต้องได้รับการรักษาทางการแพทย์เพื่อฟื้นฟูแต่ไม่ถึงขั้นหยุดงาน	<u>Level 2</u> 1 ครั้งใน 10 ปี
	<u>Level 1</u> รู้สึกไม่สะดวกสบายชั่วคราว รำคาญ หงุดหงิด ปวดเมื่อย	<u>Level 1</u> 1 ครั้งใน 20 ปี
ผลกระทบด้านการ ปฏิบัติการ (Operation)	<u>Level 4</u> Zero sendout หรือ ไม่สามารถรับเรือ/ส่งออกทางรถไฟ หรือ ไม่ปฏิบัติตามกฎหมาย	<u>Level 4</u> >1 ครั้งต่อปี
	<u>Level 3</u> ทำให้ Availability ของการส่งออกหรือการรับเรือ ลดลง หรือ ไม่ปฏิบัติตามกฎระเบียบของหน่วยงานภายนอกที่กำกับดูแล	<u>Level 3</u> 1 ครั้งใน 3 ปี
	<u>Level 2</u> หยุดต่อเนื่อง นานกว่า 48 ชั่วโมง หรือ ไม่ปฏิบัติตามกฎระเบียบของบริษัท	<u>Level 2</u> 1 ครั้งใน 5 ปี
	<u>Level 1</u> ไม่มีผลกระทบต่อการทำงาน หรือ ไม่เป็นไปตามมาตรฐานระบบงานที่เกี่ยวข้องของบริษัท	<u>Level 1</u> 1 ครั้งใน 10 ปี
ผลกระทบด้านพลังงาน (Energy)	<u>Level 4</u> Loss ค่าใช้จ่าย > 2 ล้านบาท/ปี	<u>Level 4</u> ต่อเนื่อง
	<u>Level 3</u> Loss ค่าใช้จ่าย 1 - 2 ล้านบาท/ปี	<u>Level 3</u> 150-300 ชั่วโมง/เดือน
	<u>Level 2</u> Loss ค่าใช้จ่าย 0.5 - 1 ล้านบาท/ปี	<u>Level 2</u> 30-150 ชั่วโมง/เดือน
	<u>Level 1</u> Loss ค่าใช้จ่าย < 0.5 ล้านบาท/ปี	<u>Level 1</u> x < 30 ชั่วโมง/เดือน
ผลกระทบด้านชื่อเสียง องค์กรและสิ่งแวดล้อม (Corporate reputation and Environmental)	<u>Level 4</u> กระทบรุนแรง	<u>Level 4</u> >1 ครั้งต่อปี
	<u>Level 3</u> กระทบมาก	<u>Level 3</u> 1 ครั้งใน 3 ปี
	<u>Level 2</u> กระทบปานกลาง	<u>Level 2</u> 1 ครั้งใน 5 ปี
	<u>Level 1</u> กระทบน้อย	<u>Level 1</u> 1 ครั้งใน 10 ปี
ผลกระทบด้านการเงิน (Finance)	<u>Level 4</u> ค่าใช้จ่าย >10 ล้านบาท	<u>Level 4</u> >1 ครั้งต่อปี
	<u>Level 3</u> ค่าใช้จ่าย 5-10 ล้านบาท	<u>Level 3</u> 1 ครั้งใน 3 ปี
	<u>Level 2</u> ค่าใช้จ่าย 2.5-5 ล้านบาท	<u>Level 2</u> 1 ครั้งใน 5 ปี
	<u>Level 1</u> ค่าใช้จ่าย < 2.5 ล้านบาท	<u>Level 1</u> 1 ครั้งใน 10 ปี





Priority Classification

Risk Matrix				
Consequences	Likelihood			
	ระดับ 1	ระดับ 2	ระดับ 3	ระดับ 4
ระดับ 1	D	C	C	B
ระดับ 2	C	C	B	B
ระดับ 3	C	B	B	A
ระดับ 4	B	B	A	A

Priority Class : A

ผลกระทบด้านพลังงาน (Energy)	<u>Level 4</u> Loss ค่าใช้จ่าย > 2 ล้านบาท/ปี	<u>Level 4</u> ต่อเนื่อง
	<u>Level 3</u> Loss ค่าใช้จ่าย 1 - 2 ล้านบาท/ปี	<u>Level 3</u> 150-300 ชั่วโมง/เดือน
	<u>Level 2</u> Loss ค่าใช้จ่าย 0.5 - 1 ล้านบาท/ปี	<u>Level 2</u> 30-150 ชั่วโมง/เดือน
	<u>Level 1</u> Loss ค่าใช้จ่าย < 0.5 ล้านบาท/ปี	<u>Level 1</u> x < 30 ชั่วโมง/เดือน





Risk Assessment

Question	Answer (Yes, No, N/A)	Remarks
1. If the modification affects any ventilation system, will modified ventilation system comply with appropriate codes and standards?	N/A	
2. Does the modification alter the normal permitted discharges to the air or water of the facility? If yes, are any changes needed to existing permits?	N/A	
3. Does the modification require that the onsite inventory of hazardous chemicals be increased? If yes, are any additional safeguards or controls needed?	N/A	
4. Does the modification require that hazardous chemicals be stored at higher temperatures or pressures than the current configuration?	N/A	
5. Does the modification require that hazardous chemical detectors be added?	N/A	
6. Are the indications and alarms in the existing plant adequate for the modified plant?	N/A	PHA Workshop required.
7. Does the modification require additional personnel protection measures (e.g., PPE)?	N/A	Restrict area shall be wear PPE 100%
8. Have the modifications created any additional vulnerabilities to external events which did not exist before (vehicle traffic, fire, water accumulation, high wind, etc.)?	N/A	
9. Is electrical power and other services adequate to meet the demands of the modified plant?	Yes	motor pumps are existing designs
10. Are the existing site security procedures adequate?	N/A	- Control work permit before access ISB building - Use access control system
11. Are the existing eyewash/safety showers adequate?	N/A	
12. Has the change been reviewed and approved by appropriate facility management?	N/A	
13. Have all other health and safety issues been addressed?	N/A	
14. Is the existing fire suppression system adequate for the change in chemical inventory?	N/A	





Node 1: VSD Speed Control & Logic				
Control	What if VSD runs at very LOW speed?	Motor Overheating: In-tank pumps rely on LNG flow through the motor for cooling/lubrication and Low flow	- kick back Pump (Min Flow)	
			-Program Speed Limit in VSD parameters 88-100%.	
Control	What if VSD runs ABOVE design speed (> 50/60Hz)?	Motor overload inside tank (difficult replacement).	-Motor Overload Relay	
			-Program Speed Limit in VSD parameters 88-100%.	
Control	What if VSD operates at a "Critical Speed"?	High vibration leading to lower bearing failure or column pipe fatigue.	-CPMS vibration monitoring	Identify resonant frequencies and program "Skip Frequencies" by Nikkiso vendor
			-Program Speed Limit in VSD parameters 88-100%.	
Node 2: Electrical & Submerged Motor				
Electrical	What if dV/dt (Voltage Spikes) are high?	High voltage spikes degrade submerged motor windings, leading to short circuit inside LNG tank.	-Install Output Sine Wave Filter. Verify motor cable length limits. (Add in VSD data sheet)	
			-Health check with PdMA (Routine PM MT.Elec)	
			-Temperature winding temp at CPMS monitor and alarm.	
Electrical	What if Loss of VSD Low-Voltage Control Power Supply (e.g., dedicated control transformer fuse blows, rectifier fault, or UPS battery depletion) or catastrophic failure of a single VSD power cell (in MV configurations).	Loss of control logic integrity or inability to maintain required voltage phasing/waveform. The VSD detects an undervoltage or imbalance fault (e.g., F001 or F009) and initiates an immediate protective trip, halting the LNG pump. This instantaneous power cessation results in zero flow.	Uninterruptible Power Supply (UPS) backup for critical VSD control logic boards.	
			Power cell bypass/isolation mechanisms (for modular medium-voltage VSDs).	
			Blower fan redundant.	
Node 3: Process Interaction (LNG)				
Process	What if ramp-up rate is too fast?	Pressure surge in discharge piping.	-Manual open kick back valve according to operation	
			-Ramp rate has set 20 seconds	
			-High pressure discharge pump trip.	
Process	What if Lower VSD speed results in insufficient generated pressure head, failing to maintain the required discharge pressure.	Insufficient pump discharge pressure leads to inadequate LNG flow rate to the recondenser	-Recondenser is designed to have sufficient volume buffer to handle in case of one in-tank pump trip.	
			-Low Discharge Pressure Trip interlock that trips the pump	
			-Low Flow Alarm on the main send-out line;	





Scope and preliminary design

1. ติดตั้ง VSD 2 units สำหรับ LP pump 2 units
2. งานลากสาย และเข้าสาย MV cable, LV cable และ Control cable
3. งานต่อเติมชัฟพอร์ทสำหรับอุปกรณ์ VSD
4. ปรับปรุงแก้ไขระบบ DCS เพื่อใช้ควบคุม VSD
5. ปรับปรุงแก้ไขระบบ CSCS เพื่อ monitor status อุปกรณ์ VSD

VSD 2 unit		Baht
Power cable & control cable		Baht
Terminator MV		Baht
Accessories		Baht
DCS modify		Baht
CSCS modify		Baht
Construction and Test		Baht
Management, SAT and training		Baht

Total

Baht



Budget

MTHB





Project Schedule

Action Plan	2025						2026											
	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
<u>MOC Authorization Phase</u>																		
1. Prepare MOC Package and Engineering design																		
2. MOC Small group approve																		
3. MOC Committee approve																		
4. TOR and Procurement																		
<u>Execute phase</u>																		
1.VSD Fabrication and local material fabrication																		
2. Site installation																		
3. MOC Closeout																		

Actual Plan





Thank you.

Q&A